

Java Collections and Frameworks

LOGGER

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and the user's name (KARNATI UDAY KIRAN REDDY). The left sidebar contains a 'QUESTIONS' section with '8 / 8 completed' and a dropdown menu set to 'Logger'. Below this is a 'LOGGER' section with instructions: 'You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class called "Log4jLogger" that uses the Log4j'. The main 'PROGRAM EDITOR' area shows the following Java code:

```
1 interface Logger {
2     void debug(String message);
3     void info(String message);
4     void warning(String message);
5     void error(String message);
6 }
7
8 public class Log4jLogger implements Logger {
9
10    @Override
11    public void debug(String message) {
12        System.out.println("DEBUG: " + message);
13    }
14
15    @Override
16    public void info(String message) {
17        System.out.println("INFO: " + message);
18    }
19
20    @Override
21    public void warning(String message) {
```

The 'OUTPUT' section shows the results of running the code:

```
DEBUG: This is a debug message
INFO: This is an info message
WARNING: This is a warning message
ERROR: This is an error message
```

The 'INPUT' section on the right contains the text: 'Your INPUT goes here!'

SORT

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and the user's name (KARNATI UDAY KIRAN REDDY). The left sidebar contains a 'QUESTIONS' section with '8 / 8 completed' and a dropdown menu set to 'Sort'. Below this is a 'SORT' section with instructions: 'Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.' The main 'PROGRAM EDITOR' area shows the following Java code:

```
1 import java.util.ArrayList;
2 import java.util.Collections;
3 import java.util.List;
4 public class Main {
5     public static void main(String[] args) {
6         List<Integer> numbers = new ArrayList<>();
7         numbers.add(45);
8         numbers.add(12);
9         numbers.add(78);
10        numbers.add(3);
11        numbers.add(29);
12        System.out.println("Original list: " + numbers);
13        Collections.sort(numbers);
14        System.out.println("Sorted list : " + numbers);
15    }
16 }
```

The 'OUTPUT' section shows the results of running the code:

```
Original list: [45, 12, 78, 3, 29]
Sorted list : [3, 12, 29, 45, 78]
```

The 'INPUT' section on the right contains the text: 'Your INPUT goes here!'

MAX-MIN



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JAVA PROGRAMMING

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QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.*;
2
3 public class MaxMinSet {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6         Set<Integer> numbers = new HashSet<>();
7
8         System.out.println("Enter the number of elements:");
9         int n = scanner.nextInt();
10
11         System.out.println("Enter " + n + " integers:");
12         for (int i = 0; i < n; i++) {
13             numbers.add(scanner.nextInt());
14         }
15
16         if (numbers.isEmpty()) {
17             System.out.println("The set is empty.");
18         } else {
19             int max = Collections.max(numbers);
20             int min = Collections.min(numbers);
21         }
22     }
23 }
```

OUTPUT

```
Enter the number of elements:
Enter 5 integers:
Set elements: [1, 2, 3, 4, 5]
Maximum value: 5
Minimum value: 1
```

INPUT

```
5
1
2
3
4
5
```

LINKED HASHMAP



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QUESTIONS 8 / 8 completed

LinkedHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.LinkedHashMap;
2 import java.util.Map;
3
4 public class Main {
5     public static void main(String[] args) {
6         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
7         map.put("Apple", 10);
8         map.put("Banana", 20);
9         map.put("Mango", 30);
10        map.put("Orange", 40);
11
12        System.out.println("Keys in insertion order:");
13        for (String key : map.keySet()) {
14            System.out.println(key);
15        }
16    }
17 }
```

OUTPUT

```
Keys in insertion order:
Apple
Banana
Mango
Orange
```

INPUT

```
Your INPUT goes
here!
```

PRIORITYQUEUE

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QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE
Write a Java program to create a PriorityQueue of integers and add elements to it.**PROGRAM EDITOR** Java Run Save**OUTPUT****INPUT**
Your INPUT goes here!

TREE MAP

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QUESTIONS 8 / 8 completed

TreeMap

TREEMAP
Write a Java program to create a TreeMap of strings and integers and iterate over its entries.**PROGRAM EDITOR** Java Run Save**OUTPUT****INPUT**
Your INPUT goes here!

HASHSET

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QUESTIONS 8 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class hash
3 {
4     public static void main(String[] args)
5     {
6         Set<String>a=new HashSet<String>();
7         a.add("uday");
8         a.add("kiran");
9         a.add("Reddy");
10        System.out.println(a);
11    }
12 }
```

INPUT

Your INPUT goes here!

OUTPUT

[1, 2, 3]